2.10.54

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Question

let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be unit vectors such that $\mathbf{a} + \mathbf{b} + \mathbf{c} = \mathbf{0}$. Which of the following are correct?

1. $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} = \mathbf{0}$
2. $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0}$
3. $\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{a} \times \mathbf{c} \neq \mathbf{0}$
4. $\mathbf{a} \times \mathbf{b}, \mathbf{b} \times \mathbf{c}, \mathbf{c} \times \mathbf{a}$ are mutually perpendicular.

Cross Product and Gram Matrix

Given ,

$$\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}, \|\mathbf{a}\|=\|\mathbf{b}\|=\|\mathbf{c}\|=1 \tag{1}$$

$$(\mathbf{a}+\mathbf{b}+\mathbf{c})^2=0 \tag{2}$$

$$\|\mathbf{a}\|^2+\|\mathbf{b}\|^2+\|\mathbf{c}\|^2+2\mathbf{a}^\top\mathbf{b}+2\mathbf{b}^\top\mathbf{c}+2\mathbf{c}^\top\mathbf{a}=0 \tag{3}$$

$$\mathbf{a}^\top\mathbf{b}+\mathbf{b}^\top\mathbf{c}+\mathbf{c}^\top\mathbf{a}=\frac{-3}{2} \tag{4}$$

$$\mathbf{a}+\mathbf{b}+\mathbf{c}=\mathbf{0}\implies\begin{pmatrix}\mathbf{a} & \mathbf{b} & \mathbf{c}\end{pmatrix}\begin{pmatrix}1\\1\\1\end{pmatrix}=\mathbf{0}\implies\mathbf{A}\mathbf{x}=\mathbf{0} \tag{5}$$

$$\mathbf{A}^\top\mathbf{A}\mathbf{x}=\mathbf{0}\implies G\mathbf{x}=\mathbf{0} \tag{6}$$

$$\begin{pmatrix} \|\mathbf{a}\|^2 & \mathbf{a}^\top \mathbf{b} & \mathbf{c}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{b} & \|\mathbf{b}\|^2 & \mathbf{b}^\top \mathbf{c} \\ \mathbf{c}^\top \mathbf{a} & \mathbf{b}^\top \mathbf{c} & \|\mathbf{c}\|^2 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \mathbf{0} \tag{7}$$

$$\begin{pmatrix} 1 + \mathbf{a}^\top \mathbf{b} + \mathbf{c}^\top \mathbf{a} \\ \mathbf{a}^\top \mathbf{b} + 1 + \mathbf{b}^\top \mathbf{c} \\ \mathbf{c}^\top \mathbf{a} + \mathbf{b}^\top \mathbf{c} + 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \tag{8}$$

$$1 + \mathbf{a}^\top \mathbf{b} + \mathbf{c}^\top \mathbf{a} = \mathbf{a}^\top \mathbf{b} + 1 + \mathbf{b}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{a} + \mathbf{b}^\top \mathbf{c} + 1 \tag{9}$$

$$\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{a} \tag{10}$$

From (4.4)

$$\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{a} = \frac{-1}{2} \tag{11}$$

$$\mathbf{a}^\top \mathbf{b} = \mathbf{b}^\top \mathbf{c} = \mathbf{c}^\top \mathbf{a} \implies \mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \tag{12}$$

$$\mathbf{a} \times \mathbf{b} = \mathbf{b} \times \mathbf{c} = \mathbf{c} \times \mathbf{a} \neq \mathbf{0} \tag{13}$$

C Code

```c
#include <stdio.h>

// Function to compute cross product of two vectors
void cross(double u[3], double v[3], double result[3]) {
    result[0] = u[1]*v[2] - u[2]*v[1];
    result[1] = u[2]*v[0] - u[0]*v[2];
    result[2] = u[0]*v[1] - u[1]*v[0];
}

int main() {
    // Example unit vectors at 120Â° apart in the xy-plane
    double a[3] = {1.0, 0.0, 0.0};
    double b[3] = {-0.5, 0.86602540378, 0.0}; // cos120, sin120
    double c[3] = {-0.5, -0.86602540378, 0.0}; // cos240, sin240

    double ab[3], bc[3], ca[3];
```

```
    cross(a, b, ab);
    cross(b, c, bc);
    cross(c, a, ca);

    printf("a Ã b = (%lf, %lf, %lf)\n", ab[0], ab[1], ab[2]);
    printf("b Ã c = (%lf, %lf, %lf)\n", bc[0], bc[1], bc[2]);
    printf("c Ã a = (%lf, %lf, %lf)\n", ca[0], ca[1], ca[2]);

    return 0;
}
```

Python Code 1

```
    import ctypes
import math

# Load shared library
lib = ctypes.CDLL("./libcross.so")

# Define function signature
lib.compute_crosses.argtypes = [
    ctypes.POINTER(ctypes.c_double), # a
    ctypes.POINTER(ctypes.c_double), # b
    ctypes.POINTER(ctypes.c_double), # c
    ctypes.POINTER((ctypes.c_double*3)*3) # result[3][3]
]

# Define the vectors (unit vectors, 120Â° apart in xy-plane)
a = (ctypes.c_double * 3)(1.0, 0.0, 0.0)
b = (ctypes.c_double * 3)(-0.5, math.sqrt(3)/2, 0.0)
c = (ctypes.c_double * 3)(-0.5, -math.sqrt(3)/2, 0.0)
```

```
# Prepare result storage (3x3 array of doubles)
ResultArray = (ctypes.c_double * 3) * 3
result = ResultArray()

# Call the C function
lib.compute_crosses(a, b, c, result)

# Print results
print("a Ã b =", [result[0][i] for i in range(3)])
print("b Ã c =", [result[1][i] for i in range(3)])
print("c Ã a =", [result[2][i] for i in range(3)])
```

Python Code 2

```
import matplotlib.pyplot as plt
import numpy as np
from mpl_toolkits.mplot3d import Axes3D

# Define the vectors (unit vectors 120Â° apart in xy-plane)
a = np.array([1.0, 0.0, 0.0])
b = np.array([-0.5, np.sqrt(3)/2, 0.0])
c = np.array([-0.5, -np.sqrt(3)/2, 0.0])

# Cross product (all same)
ab = np.cross(a, b)

# Create 3D plot
fig = plt.figure(figsize=(7,7))
ax = fig.add_subplot(111, projection="3d")
# Plot vector a (red arrow)
ax.quiver(0,0,0, *a, color="r")
```

```
ax.text(a[0], a[1], a[2], "a", color="r")
# Plot vector b (green thick line)
ax.plot([0, b[0]], [0, b[1]], [0, b[2]], color="g", linewidth=3)
ax.text(b[0] + 0.2, b[1], b[2], "b", color="g") # label moved right
# Plot vector c (blue arrow)
ax.quiver(0,0,0, *c, color="b")
ax.text(c[0] - 0.2, c[1], c[2], "c", color="b") # label moved left
# Plot cross product vector (black arrow)
ax.quiver(0,0,0, *ab, color="k", linestyle="dashed")
ax.text(ab[0], ab[1], ab[2], "aÃb = bÃc = cÃa", color="k")

# Axes settings
ax.set_xlabel("X")
ax.set_ylabel("Y")
ax.set_zlabel("Z")
ax.set_xlim([-1.5, 1.5])
ax.set_ylim([-1.5, 1.5])
ax.set_zlim([0, 1.2])
ax.set_title("Vectors a, b, c and Cross Products")
```

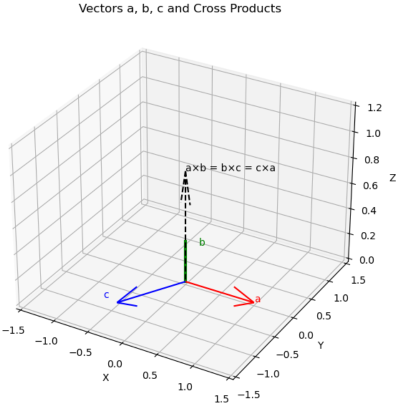

Figure: fig : Representation of vectors